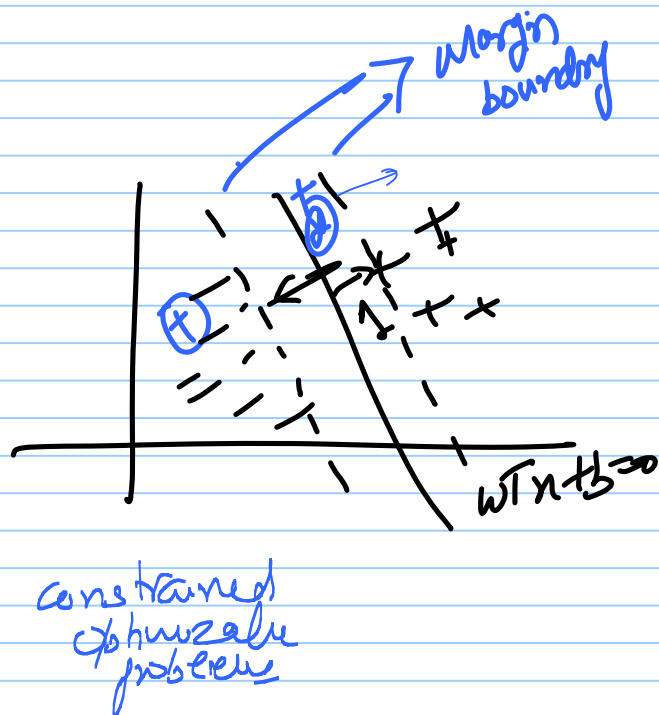




COL774
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SVMs:- Soft-margin SVM.

Objective: maximize the min. distance
by parameter

$$\min_{w, b} \frac{1}{2} w^T w + C \sum_{i=1}^n \xi_i$$

yes $(w^T x^{(i)} + b) \geq 1 - \xi_i$

Hard margin formula

$\xi_i \geq 0$
soft margin $\rightarrow$ Hard margin

Theory of Lagrangian:- (Primal - Dual problem)

$$\{\alpha_i\}_{i=1}^m$$

$$\{\gamma_i\}_{i=1}^m$$

Primal $\mathcal{L}(w, b, \epsilon, \alpha, \gamma) =$

w $\rightarrow$ Langrangian $\rightarrow$ Dual variables

$$\left[\frac{1}{2} w^T w + c \sum \epsilon_i \right] + \sum_{i=1}^m \alpha_i [1 - \epsilon_i - \gamma_i / (w^T x_i + b)]$$

$$+ \sum_{i=1}^m \gamma_i (-\epsilon_i)$$

$$[\alpha \geq 0, \gamma \geq 0]$$

$w \in \mathbb{R}^n$

$\rightarrow$ $\min_w f(w)$ $\rightarrow$ convex

$L \in \{1, \dots, m\}$ $\rightarrow$ convex $\epsilon g_i(w) \leq 0$

$l \in \{1, \dots, k\}$ $\rightarrow$ linear/affine $+ h_l(w) \geq 0$

$$Q_p(w) =$$

$\rightarrow$ Langrangian

$$\mathcal{L}(w, \alpha, \beta) = f(w) + \sum_i \alpha_i g_i(w) + \sum_l \beta_l h_l(w)$$

$$\alpha \geq 0$$

$$+ \left[\max_{\alpha \geq 0, \beta} \mathcal{L}(w, \alpha, \beta) \right]$$

if constraints are satisfied

$$= \left[\max_{\alpha \geq 0, \beta} \mathcal{L}(w, \alpha, \beta) \right] = f(w)$$

if constraints are not satisfied

$$\rightarrow \infty$$

Primal Objective

Original problem = $\left[\min_w Q_p(w) \right]$

(Primal)

$$\max_{\substack{\alpha, \beta \\ \alpha \geq 0}} Q_D(\alpha, \beta) = \max_{\substack{\alpha, \beta \\ \alpha \geq 0}} \left[\min_w L(w, \alpha, \beta) \right] \quad (\text{Dual})$$

$$Q_D(\alpha, \beta) = \min_w L(w, \alpha, \beta)$$

$$\forall w, \alpha, \beta: -Q_P(w) \geq Q_D(\alpha, \beta)$$

~~Primal~~ $\Rightarrow$ (duality gap) $\min_w Q_P(w) \geq \max_{\substack{\alpha, \beta \\ \alpha \geq 0}} Q_D(\alpha, \beta)$

SLM: Primal Objective $= \left[\max_{\substack{\alpha, \gamma \\ \alpha \geq 0, \gamma \geq 0}} L(w, b, c, \alpha, \gamma) \right]$

$$\min_{w, b, c} Q_P(w, b, c) = \min_{w, b, c} \left[\max_{\substack{\alpha \geq 0 \\ \gamma \geq 0}} L(w, b, c, \alpha, \gamma) \right]$$

$$\max_{\substack{\alpha \geq 0 \\ \gamma \geq 0}} Q_D(\alpha, \gamma) = \max_{\substack{\alpha \geq 0 \\ \gamma \geq 0}} \left[\min_{w, b, c} L(w, b, c, \alpha, \gamma) \right]$$